INTERACTIVE DASHBOARD CREATION

USING

Data Studio

A Complete Step-by-Step Lecture Guide

HR Attrition Analytics Dashboard | IBM Dataset | 1,470 Employees | 32 Fields

Two-Page Dashboard: Page 1 — Basic | Page 2 — Advanced (Calculated Fields)

1,470	16.1%	2	32	Free
Employees in the Dataset	Attrition Rate	Dashboard Pages	Data Fields	Data Studio Tool

Dataset: HR Analytics Final Dataset.xlsx

Create a copy of the dataset to use it: **Go to File→ Save as Google Sheet**

Section 1: Introduction to Data Studio

What is Data Studio?

Data Studio (formerly Google Data Studio) is a FREE, browser-based data visualization and reporting tool by Google. You do not need to install anything — it runs entirely in your web browser. With Data Studio, you can connect to data sources like Google Sheets, Excel files, databases, and more, then drag and drop charts, tables, and filters to build beautiful, interactive dashboards that automatically update when your data changes.

Why Data Studio for this session?

- Completely free — no subscription, no credit card, no download needed
- Works in any browser (Chrome recommended) — just go to lookerstudio.google.com
- Beginner-friendly drag-and-drop interface — no coding required for basic dashboards
- Supports calculated fields for advanced analysis (what we'll do on Page 2)
- Shareable with a link — just like a Google Doc, anyone with the link can view it
- Industry-standard tool used by marketing teams, HR analysts, and business intelligence teams worldwide

What we will build today:

An HR Attrition Analytics Dashboard for IBM — a company analyzing why its employees are leaving. Using real IBM HR data (1,470 employees, 32 fields), we will build a professional 2-page dashboard that any HR manager could use to make data-driven decisions.

Page 1 — The Overview Dashboard: KPI cards, charts, and filters using raw data fields. No formulas needed.

Page 2 — The Advanced Analysis Dashboard: Uses Data Studio's Calculated Fields to create new metrics like Attrition Rate %, income brackets, and performance categories that don't exist in the raw data.

1.1 Before You Begin — What You Need

Requirement	Details	Where to Get It
Google Account	A Gmail or Google Workspace account (any @gmail.com works)	Sign up free at accounts.google.com if you don't have one
Web Browser	Google Chrome is strongly recommended for best compatibility	Download at google.com/chrome if not installed
The Dataset	HR_Analytics_Final_Dataset.xlsx — the file used in this session	Download from the session materials shared by your instructor

Requirement	Details	Where to Get It
Google Sheets	We'll upload the Excel file to Google Sheets as the data source	Built into your Google Account — no separate download needed
Internet Connection	Required — Data Studio is a cloud tool; everything is online	Use a stable WiFi or wired connection for the session

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IMPORTANT — Data Studio requires a Google Account

If you don't have a Google Account, please create one NOW before continuing. Go to accounts.google.com → click 'Create account'. Everything else in this guide assumes you are logged in to Google.

Section 2: Preparing Your Data (Google Sheets Upload)

Why Google Sheets?

Data Studio connects most easily to Google Sheets. While it can connect to Excel directly, converting to Google Sheets gives us the most reliable, error-free connection with full feature support. This two-minute upload step saves hours of connection troubleshooting.

STEP 01

Upload the Excel File to Google Sheets

Follow these exact steps in order:

1. Open a new browser tab and go to: sheets.google.com
2. Sign in with your Google account if prompted
3. In the top-left corner, click the large '+' button (it says 'Blank spreadsheet' on hover)
4. A new blank spreadsheet opens. Now go to: File menu → Import
5. In the Import dialog box, click the 'Upload' tab at the top
6. Click 'Browse' or drag your HR_Analytics_Final_Dataset.xlsx file into the upload area
7. Wait for the file to upload (you'll see a progress bar)
8. An 'Import file' dialog appears. Set these options exactly:

Setting	What to Select	Why This Matters
Import location	Replace spreadsheet	Replaces the blank sheet v opened — keeps things cl
Separator type	Auto-detect (usually pre-selected)	Excel files handle this automatically
Convert text to numbers and dates	Yes (keep this checked)	Ensures Data Studio reads numbers as numbers, not

9. Click 'Import data' button
10. The spreadsheet loads with all 1,470 rows and 32 columns — you can see the data
11. Rename the sheet: click the tab at the bottom (might say 'main' or 'Sheet1') → rename it to 'HR Data' for clarity
12. Rename the file at the top: click the filename area → rename it to 'IBM HR Analytics Dataset'
13. CRITICAL: Click anywhere in the data to check Row 1 has your column headers (EmployeeNumber, Age, Attrition, etc.) and Row 2 starts with the first data record (employee number 1)

How to verify your upload worked correctly:

- Row 1 should be: EmployeeNumber | Age | Attrition | BusinessTravel (column headers)
- Row 2 should be: 1 | 41 | Yes | Travel_Rarely | ... (first employee's dat
- The sheet should have exactly 1,471 rows total (1 header + 1,470 employees)
- Column count: 32 columns (A through AF)

STEP
02

Verify Critical Columns Before Connecting

Before connecting to Data Studio, quickly verify these key columns exist and look correct:

Column Name	Should Contain	Example Values
Attrition	Text: 'Yes' or 'No'	Yes, No, No, Yes, No...
Department	Text: department names	Sales, Research & Development, Resources
MonthlyIncome	Numbers: 1,009 to 19,999	5993, 5130, 2090, 2909...
Age	Numbers: 18 to 60	41, 49, 37, 33, 27...
JobRole	Text: job title	Sales Executive, Research Scien Technician...
OverTime	Text: 'Yes' or 'No'	Yes, No, Yes, Yes, No...
Gender	Text: 'Male' or 'Female'	Female, Male, Male, Female, Mal
JobSatisfaction	Numbers: 1 to 4	4, 2, 3, 3, 2...

If any column looks wrong (all zeros, blank, 'N/A'):

Go back to File → Import and repeat Step 01, making sure 'Convert text to numbers dates' is checked. A wrong import is the most common beginner mistake.

Section 3: Setting Up Data Studio & Connecting Your Data

STEP 03

Open Data Studio and Create a New Report

14. Open a new browser tab and navigate to: https://datastudio.google.com/u/0/navigation/reporting
15. You will land on the Data Studio home page. Make sure you are signed in (your Google profile photo should appear in the top-right corner)
16. Click the large blue button: '+ Create' or 'Blank Report' (the exact label depends on your screen)
17. A dialog box appears: 'Welcome to Data Studio! Add data to report'
18. This is the data connection screen — we'll set it up in Step 04

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What you see on the Data Studio Home Page:

- Recent reports (empty for new users)
- Templates Gallery — pre-built report templates you can explore later
- + Create button — this is what we click to start fresh
- Top navigation: Home | Reports | Data Sources | Explorer

STEP 04

Connect Google Sheets as Your Data Source

After clicking 'Blank Report', the 'Add data to report' panel appears on the right side. Follow these steps:

19. In the data connector list, scroll to find 'Google Sheets' — it has a green spreadsheet icon
20. Click 'Google Sheets' — a new panel opens showing your Google Drive files
21. Under 'Spreadsheet', find and click 'IBM HR Analytics Dataset' (the file we uploaded in Step 01)
22. Under 'Worksheet', select 'HR Data' (the sheet tab we named)
23. At the bottom, you will see a preview showing your column names — verify they look correct
24. Important: ensure 'Use first row as headers' is CHECKED (it usually is by default)
25. Click the blue 'Add' button in the bottom-right
26. A confirmation dialog says 'You are about to add data to this report' — click 'Add to report'

A blank white canvas appears — this is your Data Studio report! You have successfully connected your data.

STEP 05

Understanding the Data Studio Interface

Take a moment to familiarise yourself with the interface before building:

Interface Area	Location on Screen	What It Does
Menu Bar	Very top of screen	File, View, Insert, Resource, Help — like any off
Toolbar	Below menu bar	Add chart, add filter, undo/redo, zoom controls, settings
Canvas	Large white area in the middle	The actual dashboard you're building — drag th here
Data Panel (right side)	Right sidebar	Shows all your data fields grouped by type (Dim = text/category, Metrics = numbers)
Properties Panel (right side)	Appears when you click a chart	Controls what data the selected chart shows
Pages Panel (left side)	Left sidebar	Shows all pages in your report — we'll have Pag Page 2

Two types of fields you'll work with constantly:

Dimensions (shown with a green ABC icon): Text or categorical fields like Departme Gender, JobRole, Attrition. Used for X-axis labels, chart categories, filter options.

Metrics (shown with a blue # icon): Numeric fields like MonthlyIncome, Age, Count o Records. Used for Y-axis values, KPI card numbers, chart heights.

STEP 06

Set Up the Report Theme and Page Size

Before adding any charts, set up the visual foundation:

Set the Theme:

27. Go to: Theme and layout (the paintbrush icon in the toolbar, or View menu → Theme and layout)
28. A panel opens on the right. Under 'Report settings', look for 'Canvas size'
29. Select 'Widescreen (16:9)' — this is the most professional layout for dashboards
30. Under 'Theme', you can select a color scheme. For this session, choose 'Simple Dark' or 'Navigator' for a professional look, OR keep 'Simple' for a clean white background
31. Click anywhere on the canvas to close the theme panel

Rename the Report:

32. At the very top-left of the screen, you'll see 'Untitled Report' in grey text
33. Click on 'Untitled Report' and type: 'IBM HR Attrition Analytics Dashboard'

34. Press Enter — the title is now saved

Rename Page 1:

35. At the bottom of the screen, you'll see a tab called 'Page 1' — right-click on it
36. Select 'Rename page' → type: 'HR Overview Dashboard' → press Enter

Section 4: Building Page 1 — HR Overview Dashboard (Basic)

What Page 1 will contain:

Page 1 is the 'at-a-glance' overview. A manager opens this page and instantly sees: How many employees do we have? What's our attrition rate? Which departments are most affected? All using only the raw data fields — no calculations or formulas needed.

Page 1 Elements:

- 4 KPI Scorecard Cards (Total Employees, Attrition Count, Average Age, Average Income)
- 1 Donut Chart (Attrition Split: Yes vs No)
- 1 Bar Chart (Attrition Count by Department)
- 1 Bar Chart (Employee Count by Job Role)
- 1 Pie Chart (Gender Distribution)
- 1 Stacked Bar Chart (Attrition by Business Travel)
- 3 Filter Controls (Department, Gender, OverTime)
- A title text box at the top

4.1 Adding the Page Title

STEP 07

Add a Title Text Box

37. In the top toolbar, click 'Insert' menu → select 'Text'
38. Your cursor changes to a crosshair. Click and drag on the canvas to draw a wide text box across the top of the page (roughly the full width, about 60px tall)
39. Double-click the text box to enter edit mode
40. Type: IBM HR Attrition Analysis — 2024 Overview
41. Select all the text you just typed (Ctrl+A)
42. In the text formatting toolbar that appears: set Font Size to 24, click Bold (B), set colour to your theme's primary colour (or dark navy blue)
43. Click outside the text box to deselect

To move and resize any element in Data Studio:

- Click once to select it (blue handles appear around the edges)
- Drag the element from its center to move it
- Drag the blue corner/edge handles to resize it
- Right-click → 'Format options' for precise position and size control

4.2 Adding KPI Scorecard Cards

Scorecard cards show a single big number — perfect for your most important KPIs. We'll create 4 cards in a row at the top of the dashboard.

STEP 08

Add KPI Card 1: Total Employee Count

44. Click 'Insert' menu → select 'Scorecard' (it has a big '#' icon)
45. Click and drag on the canvas to draw a small square/rectangle in the top-left area of the dashboard (below your title). Make it approximately 200px wide and 120px tall
46. The card appears with a 'Record Count' metric by default — this is perfect! It shows 1,470 (total number of rows = total employees)
47. Look at the right panel: the 'Data' tab is showing. Under 'Metric', you'll see 'Record Count' — leave this as is
48. Click the 'Style' tab in the right panel
49. Under 'Metric', set font size to 36 (for the big number)
50. Under 'Comparison label', type: 'Total Employees'
51. Under 'Primary metric label', you can type: 'Total Employees' to label the card

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Common Mistake: Why use 'Record Count' and not 'EmployeeNumber'?

'Record Count' counts how many rows exist = number of employees. If you use SUM EmployeeNumber, you'd get the sum of ID numbers (meaningless). Always use Reco Count to count employees.

STEP 09

Add KPI Card 2: Total Attrition Count

52. Click 'Insert' menu → select 'Scorecard' again
53. Draw it next to the first card (to its right, same size)
54. This time we need to COUNT only employees where Attrition = 'Yes'. We need a filter on this card:
55. In the right panel (Data tab), look for 'Filter' section near the bottom — click 'Add a filter'
56. Click 'Create a filter'
57. In the filter dialog: Name the filter 'Attrition Yes Only'
58. Under 'Include', select dimension: 'Attrition'
59. Condition: 'Equal to (=)'

60. Value: type exactly 'Yes' (capital Y, no spaces)
61. Click 'Save' to save the filter
62. Now this Scorecard will only count rows where Attrition = Yes → shows 237
63. Add label: 'Employees Who Left'

What you just did:
You applied a card-level filter that tells Data Studio: 'Only count records where Attritic equals Yes.' The result is 237 — the number of IBM employees who left. This is diffe from a Page-level filter (which affects everything on the page) — this filter only affect one card.

STEP 10

Add KPI Card 3: Average Monthly Income

64. Insert → Scorecard → draw the 3rd card next to the second one
65. In the right panel (Data tab), under 'Metric', click the existing metric to change it
66. Type 'MonthlyIncome' in the search box → select 'MonthlyIncome' from the dropdown
67. The aggregation type (the dropdown next to the field name) should say 'AVG' — if it doesn't, click the dropdown and select 'AVG' (Average)
68. This shows ₹6,503 — the average monthly income across all 1,470 employees
69. In the Style tab: under 'Number format', click the dropdown → select 'Currency' → select your local currency symbol
70. Add label: 'Avg Monthly Income'

STEP 11

Add KPI Card 4: Average Employee Age

71. Insert → Scorecard → draw the 4th card completing the row
72. In Data tab, change the Metric to 'Age' with aggregation 'AVG'
73. This shows 36.9 — the average age of IBM employees
74. In Style tab: under 'Number format', select 'Number' → set Decimal places to 1
75. Add label: 'Avg Employee Age'

You now have 4 KPI cards in a row:



4.3 Adding Charts

STEP 12

Add Chart 1: Donut Chart — Attrition Split (Yes vs No)



Donut Chart → Employee Attrition Split

Setting	Value to Set
Chart Type	Donut Chart
Dimension (X-Axis / Category)	Attrition (Dimension — the green ABC field)
Metric (Y-Axis / Value)	Record Count (Metric — will auto-count per category)
Style / Format	In Style tab: turn ON 'Show donut hole'. Add data labels showing % Colour 'Yes' slice Red, 'No' slice Teal/Green.

Why this chart? Instantly shows the 16.1% vs 83.9% attrition split. Any manager can se problem at a glance without reading numbers.

76. Click Insert → Chart (or click the chart icon in the toolbar)
77. A sidebar appears showing chart types. Select 'Pie' chart (then we'll switch to Donut in Style)
78. Click and drag on the canvas to place the chart (below and to the left of your KPI cards)
79. In the right panel, Data tab: set Dimension to 'Attrition' (click the dimension field, type Attrition, select it)
80. Set Metric to 'Record Count' (it may already be set — verify)
81. Click the Style tab: find 'Donut' toggle and switch it ON
82. Under 'Show data labels': turn ON, set label type to 'Percent'
83. Scroll to 'Slice colors': click 'No' slice and set it to a green/teal. Click 'Yes' slice and set it to red
84. Add a chart title: click the chart → Style tab → look for 'Chart header' or use Insert → Text to add a title above the chart

STEP 13

Add Chart 2: Bar Chart — Attrition Count by Department

Bar Chart (Vertical) → **Attrition Count by Department**

Setting	Value to Set
Chart Type	Bar Chart (Vertical)
Dimension (X-Axis / Category)	Department (the 3 departments: Sales, R&D, Human Resources)
Metric (Y-Axis / Value)	Record Count — with a FILTER: Attrition = Yes (so bars show only per department)
Style / Format	Sort bars in descending order. Show data labels on top of each bar distinct colours per department.

Why this chart? Shows which department is losing the most people. R&D has 961 employees, Sales has 446 — raw counts without rate context, which is intentional for the page.

85. Insert → Chart → select 'Bar chart' (vertical bars icon)
86. Place it on the canvas next to the Donut chart
87. In Data tab: set Dimension to 'Department'
88. Set Metric to 'Record Count'
89. Now add a filter to show only attrition = Yes: scroll down in Data tab → 'Add a filter' → 'Create a filter' → Include where Attrition = Yes → Save
90. In Style tab: turn ON 'Show data labels' to see the exact count above each bar
91. Sort: click the metric column header in the preview → sort descending

Expected bars: R&D (~133 leavers), Sales (~92 leavers), HR (~12 leavers). These a counts. On Page 2 (Advanced), we'll calculate the attrition RATE per department, wh tells a different story.

STEP 14

Add Chart 3: Horizontal Bar Chart — Employee Count by Job Role

Horizontal Bar Chart → **Employees by Job Role**

Setting	Value to Set
Chart Type	Horizontal Bar Chart
Dimension (X-Axis / Category)	JobRole (9 different job roles)

Metric (Y-Axis / Value)	Record Count (total employees per role — not filtered, shows all employees)
Style / Format	Horizontal bars work better when category names are long (like 'La Technician'). Sort descending by count. Show data labels.

Why this chart? Shows the workforce composition — Sales Executive is the largest role employees). This helps HR understand which roles have the most headcount and therefo most absolute attrition risk.

92. Insert → Chart → look for 'Horizontal bar chart' (bars going left to right)
93. Place it on the canvas
94. Dimension: JobRole | Metric: Record Count (no filter this time — shows all employees)
95. Style: Sort descending. Show data labels.
96. Expected result: Sales Executive (326), Research Scientist (292), Laboratory Technician (259)...

STEP 15

Add Chart 4: Pie Chart — Gender Distribution



Pie Chart → Gender Distribution

Setting	Value to Set
Chart Type	Pie Chart
Dimension (X-Axis / Category)	Gender (Male / Female)
Metric (Y-Axis / Value)	Record Count (count of employees per gender)
Style / Format	Two slices only. Show labels as both percentage and absolute cou for Male, Pink/Purple for Female.

Why this chart? Quick demographic snapshot. IBM has 882 Male (60%) and 588 Fema (40%) employees. Gender parity context is important for any attrition analysis.

97. Insert → Chart → Pie chart
98. Dimension: Gender | Metric: Record Count
99. In Style tab: set 'Show data labels' to show both values (percentage and count)
100. Set slice colours: Blue for Male, a contrasting colour for Female

STEP 16

Add Chart 5: Stacked Bar Chart — Attrition by Business Travel

Stacked Bar Chart → Attrition Status by Business Travel

Setting	Value to Set
Chart Type	Stacked Bar Chart
Dimension (X-Axis / Category)	BusinessTravel (Non-Travel, Travel_Rarely, Travel_Frequently)
Metric (Y-Axis / Value)	Record Count, stacked by Attrition (Yes/No) — achieved using Brea Dimension
Style / Format	100% stacked bars make proportions comparable. Green for 'No' a Red for 'Yes'.

Why this chart? Shows whether frequent travellers have higher attrition. Visually, the 'Y (red) portion is clearly larger for Travel_Frequently employees, revealing travel as an attri driver.

101. Insert → Chart → Stacked bar chart
102. Dimension: BusinessTravel | Metric: Record Count
103. IMPORTANT — look for a field called 'Breakdown Dimension' in the Data tab (it appears below the regular Dimension)
104. Set Breakdown Dimension to: Attrition — this creates the stacked Yes/No within each travel category
105. In Style tab: select '100% stacked' for easy proportion comparison

⚠ **If you don't see 'Breakdown Dimension':**
Click on the chart to select it. In the right panel, look at the Data tab. Scroll down pas Metric. The 'Breakdown Dimension' field should appear below Sort. If it's missing, cli '+' icon next to 'Dimension' — some chart types call it 'Breakdown Dimension' and otl call it 'Secondary Dimension'. Select Attrition there.

4.4 Adding Filter Controls (Interactive Slicers)

Filter controls let dashboard viewers click to filter everything on the page. When a user selects 'Sales' in the Department filter, ALL charts on the page update automatically to show only Sales data.

STEP 17

Add Filter Control 1: Department Filter

106. Click Insert → 'Filter control' (looks like a dropdown/list icon)

107. Draw it on the canvas (top-right area works well, or as a sidebar strip)

108. In the right panel (Data tab), the field 'Control field' should be set to: Department

109. If it's not already set, click the control field dropdown and select 'Department'

110. In Style tab: you can choose the display type — 'Drop-down list' (compact) or 'Fixed size list' (shows all options at once)

111. For this filter, select 'Drop-down list'

112. Data Studio will automatically show the 3 departments: Human Resources, Research & Development, Sales



How to test it: Click 'View' in the top-right to enter View mode (as a dashboard user w see it). Click the Department dropdown and select 'Sales' — ALL charts on the page show only Sales data. Click '(All)' to reset. Press Escape or click 'Edit' to go back to e mode.

STEP 18

Add Filter Control 2: Gender Filter & OverTime Filter

113. Repeat the same process for Gender: Insert → Filter control → Control field = Gender

114. For the display type: 'Fixed size list' (only 2 options: Male, Female — fits compactly)

115. Add a third filter for OverTime: Insert → Filter control → Control field = OverTime

116. Display type: Fixed size list (only Yes/No — very compact)

Your Page 1 should now look like this layout:

Area	Content
Top Strip	Title text: 'IBM HR Attrition Analysis — 2024 Overview'
Row 1 (KPI Cards)	Total Employees | Employees Who Left | Avg Monthly Income | Avg
Row 2 Left (large)	Donut Chart: Attrition Split
Row 2 Right (large)	Bar Chart: Attrition Count by Department
Row 3 Left	Horizontal Bar Chart: Employees by Job Role
Row 3 Right Top	Pie Chart: Gender Distribution
Row 3 Right Bottom	Stacked Bar: Attrition by Business Travel

Right Sidebar	Filter Controls: Department | Gender | OverTime

4.5 Formatting Page 1 — Making It Professional

STEP 19

Apply Consistent Formatting

Aligning elements:

117. Click on a chart. Hold Shift and click other charts to select multiple at once
118. Right-click → 'Align' → 'Align top edges' to make your KPI cards all the same height
119. Right-click → 'Distribute' → 'Distribute horizontally' to space cards evenly

Adding background shapes:

120. Insert → Shape → Rectangle — draw a light grey rectangle behind your KPI cards to group them visually
121. Right-click the rectangle → 'Send to back' so it sits behind the cards
122. Set the rectangle fill colour to a very light grey (or your theme's accent colour)

Chart titles:

123. Click each chart → Style tab → look for 'Chart header' section
124. Toggle 'Show header' ON → type a descriptive title for each chart
125. Example titles: 'Attrition Split (Yes vs No)', 'Who Is Leaving? By Department', 'Workforce by Role'



Professional tip: Use consistent colours across your dashboard. If bars showing 'Attri Yes' are red in one chart, make them red in all charts. This colour consistency makes dashboards much easier to read and reduces cognitive load for the viewer.

Section 5: Building Page 2 — Advanced Analysis (Calculated Fields)

What is a Calculated Field?

A Calculated Field is a new field you create using a formula — similar to a formula in Excel. The difference is that the formula is written directly in Data Studio and automatically applies to every row in your dataset. Calculated Fields allow you to create NEW metrics that don't exist in your raw data — like Attrition Rate %, Income Brackets, or Satisfaction Labels.

What Page 2 will contain:

- Calculated Field 1: Attrition Rate % — a dynamic percentage using a formula
- Calculated Field 2: Age Group — categorizes ages into bands (18-25, 26-35, etc.)
- Calculated Field 3: Income Bracket — groups monthly income into Low/Mid/High/Very High
- Calculated Field 4: Satisfaction Label — converts 1-4 number to text (Low/Medium/High/Very High)
- Charts using these calculated fields — bar charts and tables that previous page couldn't show
- A summary table comparing Attrition Rate across multiple dimensions

STEP 20

Add Page 2 to Your Report

126. At the bottom of the screen, look for the '+' icon next to your 'HR Overview Dashboard' tab
127. Click the '+' button — a new page appears called 'Page 2'
128. Right-click 'Page 2' → Rename → type: 'Advanced Attrition Analysis' → press Enter
129. Click on the 'Advanced Attrition Analysis' tab to make it the active page
130. Add a title text box at the top: 'Advanced Attrition Analysis — Calculated Field Insights'

5.1 Creating Calculated Fields

How to access Calculated Fields:

Calculated Fields are created in the Data Source. There are two ways to access them:

Method 1 (Recommended for beginners):

When a chart is selected → in the right panel (Data tab) → click on a Dimension or Metric field → at the bottom of the dropdown that appears, you'll see 'Create field' or 'New calculated field'. Click that.

Method 2 (For managing all fields):

Go to Resource menu (top menu bar) → 'Manage added data sources' → click 'Edit' next to your Google Sheets data source → click 'Add a field' button at the top-right. This opens the full Calculated Field editor.

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Before creating calculated fields — IMPORTANT:

Always go to Resource → Manage added data sources → Edit your Google Sheets source when creating calculated fields. This ensures the field is available to ALL charts on ALL pages, not just one chart. We'll use this method for all 4 calculated fields below.

Creating Calculated Field 1: Attrition Flag (Numeric)

Before we can calculate an Attrition RATE, we need to convert the text field 'Attrition' (which contains 'Yes' or 'No') into a number (1 or 0). This is our foundation field.

Calculated Field: Attrition Flag

```
CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END
```

What this formula does:

This formula checks each row: if the Attrition column says 'Yes', it assigns 1. If it says 'No', it assigns 0. This creates a numeric 1/0 version of attrition that we can then SUM (to count leavers) or AVERAGE (to get attrition rate).

Expected result: Each row now has either 1 (employee left) or 0 (employee stayed). SUM of this field = 237 (total attrition). AVG of this field = 0.161 = 16.1% attrition rate.

STEP 21

Step-by-Step: Create the Attrition Flag Field

131. Go to: Resource menu → 'Manage added data sources'
132. Find your 'IBM HR Analytics Dataset' data source → click 'Edit'
133. You see all 32 existing fields listed. At the top-right, click the blue 'Add a field' button
134. A new field editor opens with two boxes: 'Field Name' (top) and 'Formula' (bottom)
135. In 'Field Name' box, type exactly: Attrition Flag
136. In the 'Formula' box (the large text area), type this formula exactly:

```
CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END
```

137. Below the formula box, Data Studio shows a 'FORMULA ERROR' or a preview. If you see a preview showing 0s and 1s — great! If you see an error, check your typing very carefully:

Common Mistake	Correct Version	Why It Fails
CASE WHEN Attrition='Yes'	CASE WHEN Attrition = 'Yes'	Missing spaces around = sign
THEN 1 ELSE 0	THEN 1 ELSE 0 END	Missing END at the close of CASE stater
'Yes' with curly quotes	'Yes' with straight apostrophes	Copy-pasting from Word can insert curly — type manually
Attrition = "Yes"	Attrition = 'Yes'	Data Studio uses single quotes for text v formulas

138. Once the formula preview shows numbers (0s and 1s), click 'Save' (top-right button)

139. You're returned to the data source editor. Your new 'Attrition Flag' field appears in the list

140. Click 'Done' (top-right) to close the data source editor

Creating Calculated Field 2: Attrition Rate %

Now that we have the numeric Attrition Flag, we can calculate the rate. This field gives us the percentage of employees who left in any segment.

Calculated Field: Attrition Rate

```
SUM(Attrition Flag) / COUNT(EmployeeNumber)
```

What this formula does:

SUM(Attrition Flag) adds up all the 1s (employees who left). COUNT(EmployeeNumber) counts all employees in the current filter context. Dividing gives the proportion. When used in a chart grouped by Department, this automatically calculates attrition rate per department.

Expected result: For the full dataset: 237 / 1470 = 0.1612 = 16.12%. When filtered to Sales only: gives Sales attrition rate. When filtered to any group: gives that group's specific attrition rate.

STEP 22

Step-by-Step: Create the Attrition Rate Field

141. Go to: Resource → Manage added data sources → Edit your data source → Add a field

142. Field Name: Attrition Rate

143. Formula:

```
SUM(Attrition Flag) / COUNT(EmployeeNumber)
```

144. If you see 'Field not found' error for 'Attrition Flag', make sure you completed Step 21 first and saved it correctly
145. The default format for this field will be a decimal (like 0.161). We need to change it to a percentage:
146. After saving the formula, look for the 'Type' dropdown next to your field in the data source editor
147. Change 'Type' to 'Percent' → this tells Data Studio to display it as 16.1% instead of 0.161
148. Click 'Save' → 'Done'



When this 'Attrition Rate' field is used as a Metric in a chart with 'Department' as the Dimension, Data Studio automatically calculates: Attrition Rate for R&D = 133/961 = Sales = 92/446 = 20.6%, HR = 12/63 = 19.0%. No additional work needed!

Creating Calculated Field 3: Age Group

This field converts the continuous Age number (18-60) into categorical age bands. This allows us to create charts showing attrition by age group.

Calculated Field: Age Group

```
CASE  WHEN Age <= 25 THEN "18-25"  WHEN Age <= 35 THEN "26-35"  WHEN Age
<= 45 THEN "36-45"  ELSE "46-60"END
```

What this formula does:

Each employee is assigned to one of 4 age bands based on their age. CASE WHEN evaluates conditions in order — the first condition that is TRUE gets applied. Ages 18-25 get '18-25', ages 26-35 get '26-35', and so on. The ELSE catches everyone over 45.

Expected result: Each of the 1,470 employees gets an age group label. Young employees (18-25) show 35.8% attrition — the highest of any group. This insight is invisible if you only look at average age.

STEP 23

Step-by-Step: Create the Age Group Field

149. Resource → Manage added data sources → Edit → Add a field
150. Field Name: Age Group

151. Formula (type it exactly as shown — mind the line breaks, they're optional but help readability):

```
CASE
  WHEN Age <= 25 THEN "18-25"
  WHEN Age <= 35 THEN "26-35"
  WHEN Age <= 45 THEN "36-45"
  ELSE "46-60"
END
```

⚠ **Formula type on keyboard — do NOT copy-paste from this document:**
Type CASE, press Enter for a new line, type WHEN, etc. Copying from Word or PDF introduce hidden characters that break formulas. Always type Data Studio formulas manually in the formula editor.

152. Verify the formula preview shows values like '26-35', '36-45', etc.
153. Click Save → Done

Sort Order Issue — Important!
By default, Data Studio will sort '18-25', '26-35', '36-45', '46-60' alphabetically — which happens to be correct in this case because the numbers start with 1, 2, 3, 4 in order. If your groups were '6-10', '11-20' etc., alphabetical sorting would put '11-20' before '6-10'. For this dataset, alphabetical order works fine.

Creating Calculated Field 4: Income Bracket

Groups the MonthlyIncome (₹1,009 to ₹19,999) into 4 meaningful brackets for analysis.

Calculated Field: Income Bracket

```
CASE  WHEN MonthlyIncome < 3000 THEN "Low (< 3K)"  WHEN MonthlyIncome < 6000 THEN "Mid (3K-6K)"  WHEN MonthlyIncome < 10000 THEN "Upper-Mid (6K-10K)"  ELSE "High (10K+)"END
```

What this formula does:
Converts the continuous monthly income number into 4 labelled categories. Employees earning less than ₹3,000 are in 'Low', between ₹3,000 and ₹6,000 in 'Mid', etc. The label text includes the range for clarity on charts.

Expected result: Employees in the 'Low (<3K)' income bracket will show a much higher attrition rate than those in 'High (10K+)'. This calculated field makes that pattern visible in a chart, which would be hard to see by just looking at raw income numbers.

STEP 24

Step-by-Step: Create the Income Bracket Field

154. Resource → Manage added data sources → Edit → Add a field
155. Field Name: Income Bracket
156. Formula:

```
CASE
  WHEN MonthlyIncome < 3000 THEN "Low (< 3K)"
  WHEN MonthlyIncome < 6000 THEN "Mid (3K-6K)"
  WHEN MonthlyIncome < 10000 THEN "Upper-Mid (6K-10K)"
  ELSE "High (10K+)"
END
```

157. Save → Done



Why this is more powerful than a raw income number: You can now make a bar char 'Income Bracket' on the X-axis and 'Attrition Rate' on the Y-axis. This shows in one cl that low-income employees leave at 40%+ while high-income employees leave at les 5% — an immediately actionable HR insight.

Creating Calculated Field 5: Job Satisfaction Label

The JobSatisfaction field contains numbers 1, 2, 3, 4. This field converts them to human-readable labels.

Calculated Field: Satisfaction Level

```
CASE  WHEN JobSatisfaction = 1 THEN "1 - Low"  WHEN JobSatisfaction = 2
THEN "2 - Medium"  WHEN JobSatisfaction = 3 THEN "3 - High"  WHEN
JobSatisfaction = 4 THEN "4 - Very High"  ELSE "Unknown"END
```

What this formula does:

Converts numeric satisfaction scores to labelled text. The labels include the number (e.g., '1 - Low') so they sort in correct order AND display meaningfully. The ELSE 'Unknown' catches any unexpected values — a safety net.

Expected result: Chart axes now show 'Low / Medium / High / Very High' instead of '1 / 2 / 3 / 4'. This makes the chart immediately readable to any non-technical stakeholder without needing to look up what '3' means.

STEP

Step-by-Step: Create the Satisfaction Label Field

158. Resource → Manage added data sources → Edit → Add a field
159. Field Name: Satisfaction Level
160. Formula:

```
CASE
  WHEN JobSatisfaction = 1 THEN "1 - Low"
  WHEN JobSatisfaction = 2 THEN "2 - Medium"
  WHEN JobSatisfaction = 3 THEN "3 - High"
  WHEN JobSatisfaction = 4 THEN "4 - Very High"
  ELSE "Unknown"
END
```

161. Save → Done

Summary: You now have 5 Calculated Fields:

Field Name	Type	Formula Summary	Purpose
Attrition Flag	Number (0/1)	CASE WHEN Attrition='Yes' THEN 1 ELSE 0 END	Converts text to number for calculations
Attrition Rate	Percentage	SUM(Attrition Flag)/COUNT(EmployeeNumber)	Dynamic attrition rate for an
Age Group	Text (Category)	CASE WHEN Age<=25 THEN '18-25'...	Groups ages into 4 bands fo
Income Bracket	Text (Category)	CASE WHEN MonthlyIncome<3000 THEN 'Low'...	Groups income into 4 mean ranges
Satisfaction Level	Text (Category)	CASE WHEN JobSatisfaction=1 THEN '1-Low'...	Converts satisfaction numbe

5.2 Building Charts on Page 2 Using Calculated Fields

Now that your calculated fields are created, let's use them to build the advanced analysis charts on Page 2.

STEP 26

Chart A: Attrition Rate by Department (Using Calculated Field)

Bar Chart → Attrition RATE by Department

Setting	Value to Set
Chart Type	Bar Chart
Dimension (X-Axis / Category)	Department (same as Page 1)
Metric (Y-Axis / Value)	Attrition Rate (our calculated field — NOT Record Count)
Style / Format	Format Y-axis as %. Show data labels on bars showing the % value reference line at 16.1% (company average).

Why this chart? RATE is more meaningful than COUNT. R&D has 133 leavers (most), b 13.8% rate it's actually BETTER than Sales (20.6%). This calculated field reveals the stor raw counts hide.

162. Insert → Chart → Bar chart → place on Page 2 canvas
163. Dimension: Department
164. Metric: Attrition Rate (select the calculated field — look for the % or # icon in the field picker)
165. In Style tab: Y-axis → Number format → Percentage → 1 decimal place
166. Add chart title: 'Attrition Rate by Department (%)'
167. To add a reference line: Style tab → scroll down → 'Reference lines' → Add reference line → Fixed value → 0.161 → Label 'Company Avg: 16.1%'

Expected Result:

Department	Expected Attrition Rate	Interpretation
Sales	~20.6%	ABOVE company average — hig department
Human Resources	~19.0%	ABOVE company average — oft overlooked
Research & Development	~13.8%	BELOW company average — m

STEP 27

Chart B: Attrition Rate by Age Group

Column Chart → Attrition Rate by Age Group

Setting	Value to Set
Chart Type	Column Chart
Dimension (X-Axis / Category)	Age Group (our calculated field — shows 18-25, 26-35, 36-45, 46-6
Metric (Y-Axis / Value)	Attrition Rate (our calculated field)
Style / Format	Sort by Age Group alphabetically (which gives correct age order). U gradient colours — darkest red for highest attrition group. Show % labels.

Why this chart? Young employees (18-25) leave at 35.8% — more than DOUBLE the company average of 16.1%. This single chart makes the case for investing in early-caree retention programs.

168. Insert → Chart → Column/bar chart
169. Dimension: Age Group (the calculated field you created)
170. Metric: Attrition Rate (calculated field)
171. Sort: Sort by Dimension (Age Group) alphabetically — this gives 18-25, 26-35, 36-45, 46-60 in order
172. Style: Show data labels, format as %. Use red colour shading (darker for higher rate)

Teaching moment: Compare this chart to a chart with Dimension=Age and Metric=Re Count. The raw count chart looks completely different and doesn't clearly show which group is most at risk. The calculated Attrition Rate field makes the insight obvious.

STEP 28

Chart C: Attrition Rate by Income Bracket

Bar Chart → Attrition Rate by Income Bracket

Setting	Value to Set
Chart Type	Bar Chart
Dimension (X-Axis / Category)	Income Bracket (our calculated field — shows 4 income levels)
Metric (Y-Axis / Value)	Attrition Rate (calculated field)
Style / Format	Colour bars from red (Low income, high attrition) to green (High inc low attrition). Sort by income level in ascending order (Low → High

Why this chart? Compensation is the most actionable attrition driver. This chart shows t employees earning below ₹3,000/month leave at a much higher rate than high earners — direct input to salary review decisions.

173. Insert → Chart → Bar chart
174. Dimension: Income Bracket (calculated field)
175. Metric: Attrition Rate (calculated field)
176. Important: manually set sort order in the Data tab so bars appear Low → Mid → Upper-Mid → High (alphabetical would put 'High' first, which is confusing)
177. To fix sort order: in Data tab, find 'Sort' → set to 'Dimension' → click the field and look for a 'Manual sort' or specify sort by income amount

⚠ **Sort Order Tip for Income Bracket:**
Because our labels start with 'H' (High), 'L' (Low), 'M' (Mid), 'U' (Upper-Mid), alphabet sort gives: High → Low → Mid → Upper-Mid (wrong order!). Fix: rename your labels include a number prefix: '1-Low', '2-Mid', '3-Upper-Mid', '4-High' in your calculated fie formula. Then alphabetical sort = correct income order. Go back to Resource → Edit source → edit your Income Bracket formula to add number prefixes.

STEP 29

Chart D: Attrition Rate by Satisfaction Level



Column Chart → Attrition Rate by Job Satisfaction Level

Setting	Value to Set
Chart Type	Column Chart
Dimension (X-Axis / Category)	Satisfaction Level (our calculated field — shows 1-Low, 2-Medium, 4-Very High)
Metric (Y-Axis / Value)	Attrition Rate (calculated field)

Style / Format	Use a diverging colour scheme: red for Low satisfaction, green for ' High. The number prefix in labels ensures correct sort order.

Why this chart? Proves the satisfaction-attrition link: Low satisfaction = ~22% attrition, V
High satisfaction = ~11% attrition. HR can use this to build targeted engagement program

178. Insert → Chart → Column chart
179. Dimension: Satisfaction Level (calculated field)
180. Metric: Attrition Rate (calculated field)
181. Sort: by Dimension alphabetically — labels '1-Low', '2-Medium', '3-High', '4-Very High' sort correctly because of the number prefix we included
182. Add reference line at 0.161 (company average) to show which satisfaction levels are above/below average

STEP 30

Chart E: Summary Pivot Table — Attrition Rate by Department AND Gender

This is the most advanced visual on Page 2 — a cross-tabulation table:

Pivot Table → Attrition Rate by Department & Gender

Setting	Value to Set
Chart Type	Pivot Table
Dimension (X-Axis / Category)	Row Dimension: Department | Column Dimension: Gender
Metric (Y-Axis / Value)	Attrition Rate (calculated field) — the rate appears in each Departm Gender cell
Style / Format	Apply conditional formatting (green = low rate, red = high rate) to th cells. Add row/column totals.

Why this chart? Shows whether gender attrition differences exist within each departmer
combined DEI and retention insight impossible to see in any single-dimension chart.

183. Insert → Chart → look for 'Pivot table' in the chart type list
184. Row Dimension: Department
185. Column Dimension: Gender
186. Metric: Attrition Rate (calculated field)
187. Style tab: Turn on 'Show row totals' and 'Show column totals'

188. Style → Metric → apply conditional formatting: set colour scale (green for low %, red for high %)

Pivot tables in Data Studio work exactly like pivot tables in Excel — rows, columns, a value at each intersection. The difference is that Data Studio calculates the value dynamically using your calculated field formula, so it always shows the correct attritio for each cell.

5.3 Adding Advanced Filters to Page 2

STEP 31

Add Advanced Filter Controls

Add these additional filters on Page 2 for more granular analysis:

189. Filter 1 — Job Role: Insert → Filter control → Control field = JobRole (allows viewing attrition for specific roles)

190. Filter 2 — Age Group: Insert → Filter control → Control field = Age Group (our calculated field — filters by age band)

191. Filter 3 — Income Bracket: Insert → Filter control → Control field = Income Bracket (our calculated field)

192. Filter 4 — OverTime: Insert → Filter control → Control field = OverTime

Can calculated fields be used as filter controls?

YES — once you create a calculated field (like Age Group or Income Bracket), it app the field list just like any regular field and can be used as a filter control, chart dimens metric. This is one of the most powerful features of calculated fields.

Section 6: Sharing, Publishing & Final Touches

6.1 Adding Navigation Between Pages

STEP 32

Add Page Navigation Buttons

Create buttons so users can easily move between Page 1 and Page 2:

On Page 1:

193. Insert → Text → draw a button-shaped rectangle → type: 'Go to Advanced Analysis →'
194. Select the text box → right-click → 'Add link' → select 'Go to page' → select 'Advanced Attrition Analysis'
195. Format the text box with a background colour to look like a button

On Page 2:

196. Insert → Text → draw a button → type: '← Back to Overview'
197. Add link → Go to page → 'HR Overview Dashboard'

Data Studio automatically adds page navigation arrows at the top of the report when View mode, but adding explicit buttons makes navigation much clearer for users who familiar with the tool.

6.2 Final Quality Check — Go Through This Checklist

Page 1 Checklist:

- □ Title text box present at the top
- □ All 4 KPI cards show correct numbers: 1,470 / 237 / ₹6,503 / 36.9
- □ Donut chart shows 16.1% Yes and 83.9% No split
- □ Department bar chart shows only attrition = Yes employees (237 total across all 3 bars)
- □ Job Role horizontal bar shows all 9 roles with Sales Executive at top (326)
- □ Gender pie shows Male (882) and Female (588)
- □ Stacked bar shows 3 travel categories with Yes/No stacking
- □ All 3 filter controls work (test in View mode by clicking each filter)
- □ When you click 'Sales' in the Department filter, all charts update to Sales only

Page 2 Checklist:

- ☐ Title text box present
- ☐ All 5 calculated fields exist (Resource → Manage data sources → verify the 5 fields are listed)
- ☐ Attrition Rate by Department shows %: Sales ~20.6%, HR ~19.0%, R&D ~13.8%
- ☐ Attrition Rate by Age Group shows 18-25 as highest rate (~35.8%)
- ☐ Income Bracket chart shows decreasing attrition as income increases
- ☐ Satisfaction Level chart shows low satisfaction = higher attrition
- ☐ Pivot table shows a grid with Department rows, Gender columns, and Attrition Rate %
- ☐ Calculated field filters work correctly
- ☐ Navigation buttons work between both pages

STEP 33

Share Your Report

198. In the top-right corner, click the 'Share' button (person+ icon)
199. An 'Invite people' dialog appears — you have multiple options:

Share Option	How to Use	Best For
Share with specific people	Enter email addresses → they get a link via email	Sharing with your manager, tea instructor
Get shareable link	Click 'Change to anyone with the link' → copy the URL	Sharing widely — anyone with can view
Manage access	Set Editor vs Viewer permissions	Editors can change the report; can only interact with filters
Embed	Get HTML code to embed in a website	Publishing to company intranet website
Download as PDF	File → Download → PDF	Sharing a static snapshot (no interactivity)

200. For this session: Click 'Share' → 'Get link' → 'Anyone with the link' → 'Viewer' → Copy Link
201. Share this link with your instructor to submit your completed dashboard



Your Data Studio report auto-saves every few seconds — you never need to manual press Save. The report is stored in your Google Drive under 'My Drive'. You can find by going to drive.google.com and searching for 'IBM HR Attrition Analytics Dashboar

Section 7: Troubleshooting Common Issues

Problem	Likely Cause	Exact Fix
Calculated field formula shows error	Typo in formula, wrong quotes, or missing END	Re-type the formula manually in the editor. Use single quotes 'Yes' not double quotes. Every CASE must have END. Check for missing spaces.
Chart shows no data / blank	Wrong field type or filter is too restrictive	Click the chart → check Data tab. Is a filter applied that excludes all data? Remove the filter and re-add it. Check that Dimension and Metric fields are set.
Attrition Rate shows 1.0 instead of 16.1%	Field type not set to Percentage	Resource → Manage data sources → Edit → find 'Attrition Rate' field → change Type to 'Percent'. Or click the metric in chart Data tab and change number format to %.
Filter doesn't affect a specific chart	The chart might be from a different data source, or the filter is chart-level not page-level	Make sure you added the filter as a 'Filter control' (from Insert menu) and NOT as a chart-level filter inside the chart's Data tab. Chart-level filters are private to that chart only.
Age Group or Income Bracket sorts in wrong order	Alphabetical sort puts categories in wrong sequence	Add numeric prefixes to your labels: '1-Low', '2-Mid', '3-Upper-Mid', '4-High'. Then alphabetical = correct order. Edit the calculated field formula to add these prefixes.
Data shows 'null' or missing values in some cells	Some employees may have blank values for that field	Check the raw data in Google Sheets for blank cells. You can add IFNULL() around field names: IFNULL(JobSatisfaction, 0) to replace nulls with 0.
Charts on Page 2 are missing / show error for calculated fields	Calculated fields not saved properly, or created on wrong data source	Go to Resource → Manage added data sources. Verify your 5 calculated fields appear in the list. If missing, create them again.
Report doesn't update after changing Google Sheets data	Data Studio caches data for 15 minutes by default	Click the refresh icon (two circular arrows) in the top toolbar. Or wait 15 minutes for auto-refresh.
Can't find 'Breakdown Dimension' for stacked bar chart	Must be in correct chart type mode	Make sure you selected 'Stacked bar chart' (not regular bar chart). Breakdown Dimension appears in the Data tab only for chart types that support it (stacked bar, stacked column).
Share link doesn't work for others	Access not set to 'Anyone with the link'	Share → Get link → change from 'Restricted' to 'Anyone with the link' → Viewer. Copy the new link after changing this setting.

Section 8: Quick Reference — All Dashboard Elements

8.1 Page 1 Summary — HR Overview Dashboard

Element	Type	Dimension	Metric / Field	Filter Applied?	Key Insight
Title	Text Box	—	—	—	IBM HR Attrition Analysis — 2024 Overview
KPI Card 1	Scorecard	—	Record Count	None	Shows 1,470 total employees
KPI Card 2	Scorecard	—	Record Count	Attrition = Yes	Shows 237 employees who left
KPI Card 3	Scorecard	—	AVG MonthlyIncome	None	Shows ₹6,503 average salary
KPI Card 4	Scorecard	—	AVG Age	None	Shows 36.9 average age
Donut Chart	Pie/Donut	Attrition	Record Count	None	16.1% Yes, 83.9% No split
Bar Chart 1	Bar Chart	Department	Record Count	Attrition = Yes	R&D: 133, Sales: 92, HR: 12 leavers
Bar Chart 2	Horizontal Bar	JobRole	Record Count	None	Sales Exec: 326, Research Sci: 292...
Pie Chart	Pie Chart	Gender	Record Count	None	Male: 882 (60%), Female: 588 (40%)
Stacked Bar	Stacked Bar	BusinessTravel	Record Count	Breakdown: Attrition	Travel patterns vs attrition
Filter 1	Filter Control	Department	—	—	Filters all Page 1 charts by department
Filter 2	Filter Control	Gender	—	—	Filters all Page 1 charts by gender
Filter 3	Filter Control	OverTime	—	—	Filters all Page 1 charts by overtime status

8.2 Page 2 Summary — Advanced Attrition Analysis

Element	Type	Dimension	Metric	Calculated Field?	Key Insight
Bar Chart A	Bar Chart	Department	Attrition Rate	Yes — Attrition Rate	Sales 20.6% vs R&D 13.8%

Element	Type	Dimension	Metric	Calculated Field?	Key Insight
Column Chart B	Column Chart	Age Group	Attrition Rate	Both fields calculated	18-25 = 35.8% — highest risk age
Bar Chart C	Bar Chart	Income Bracket	Attrition Rate	Both fields calculated	Low income = highest attrition
Column Chart D	Column Chart	Satisfaction Level	Attrition Rate	Both fields calculated	Low satisfaction = ~22% attrition
Pivot Table E	Pivot Table	Dept × Gender	Attrition Rate	Yes — Attrition Rate	Gender-department cross analysis
Filter 4	Filter Control	JobRole	—	No — raw field	Filter by specific job role
Filter 5	Filter Control	Age Group	—	Yes — calculated field	Filter by age band
Filter 6	Filter Control	Income Bracket	—	Yes — calculated field	Filter by income level
Filter 7	Filter Control	OverTime	—	No — raw field	Filter by overtime status

8.3 All 5 Calculated Fields — Complete Formula Reference

Field Name	Formula (Type Exactly)	Output Type	Where Used
Attrition Flag	CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END	Number (0 or 1)	Input to Attrition Rate formula
Attrition Rate	SUM(Attrition Flag) / COUNT(EmployeeNumber)	Percentage	All attrition rate charts on Page 2 + Pivot Table
Age Group	CASE WHEN Age <= 25 THEN "18-25" WHEN Age <= 35 THEN "26-35" WHEN Age <= 45 THEN "36-45" ELSE "46-60" END	Text	Chart B dimension + Filter 5
Income Bracket	CASE WHEN MonthlyIncome < 3000 THEN "1-Low" WHEN MonthlyIncome < 6000 THEN "2-Mid" WHEN MonthlyIncome < 10000 THEN "3-Upper-Mid" ELSE "4-High" END	Text	Chart C dimension + Filter 6
Satisfaction Level	CASE WHEN JobSatisfaction = 1 THEN "1 - Low" WHEN JobSatisfaction = 2 THEN "2 - Medium" WHEN JobSatisfaction = 3 THEN "3 - High" WHEN JobSatisfaction = 4 THEN "4 - Very High" ELSE "Unknown" END	Text	Chart D dimension

Section 9: Resume Snippet for Students

After completing this session, you can add the following to your resume:

You have built a real, interactive, shareable business intelligence dashboard — not a screenshot, not a static report, but a live URL that anyone can access and interact with.

9.1 Projects Section Entry

IBM HR Attrition Interactive Dashboard | Data Studio

Tools: Data Studio (Google Data Studio) | Data Source: Google Sheets | Dataset: IBM HR Analytics (1,470 employees, 32 variables)

- Designed and published a 2-page interactive HR analytics dashboard using Data Studio (free, browser-based BI tool) connected to a live Google Sheets data source
- Built Page 1 (Overview): KPI scorecard cards, donut charts, bar charts, pie charts, and 3 interactive filter controls enabling real-time data slicing by Department, Gender, and OverTime without any code
- Built Page 2 (Advanced Analysis): Created 5 calculated fields using CASE WHEN formulas — including a dynamic Attrition Rate metric, Age Group binning, Income Bracket categorisation, and Satisfaction Level labels — enabling metrics not available in the raw dataset
- Identified key attrition insights: Sales department 20.6% attrition rate vs R&D 13.8%; 18-25 age group at 35.8% attrition rate (2.2x company average); low-income employees (<₹3K/month) showing highest attrition risk
- Configured sharing and access permissions; embedded navigation between report pages for improved user experience

9.2 Technical Skills Demonstrated

Data Studio	Data Preparation	Analytics Skills
▸ KPI Scorecard cards with metrics ▸ Donut, Bar, Pie, Stacked Bar, Pivot charts ▸ Interactive Filter Controls (slicers) ▸ Card-level and page-level filters	▸ Excel to Google Sheets migration ▸ Data type verification ▸ Column header and null validation ▸ Data source connection management	▸ KPI definition and measurement ▸ Attrition rate calculation ▸ Demographic segmentation analysis ▸ Income bracket analysis

▸ Calculated Fields (CASE WHEN formulas) ▸ Chart formatting and conditional colours ▸ Reference lines on charts ▸ Report sharing and access management	▸ Field type configuration (%, number, text) ▸ Multi-sheet workbook handling	▸ Satisfaction-attrition correlation ▸ Executive dashboard design principles ▸ Cross-tabulation (pivot) analysis

9.3 Interview Talking Point

When asked: 'Tell me about a data visualization project you've built.'

"I built an interactive HR attrition analytics dashboard using Google Data Studio for IBM's HR dataset of 1,470 employees. The dashboard has two pages: a basic overview page with KPI cards, donut charts, and interactive filters that allow any non-technical user to explore the data; and an advanced analysis page where I created calculated fields using CASE WHEN formulas to derive new metrics like Attrition Rate % by department, Age Group binning, and Income Bracket categories. The key finding was that the Sales department had a 20.6% attrition rate vs Research & Development's 13.8% — raw headcount numbers had made R&D look worse, but once I calculated the rate, Sales was clearly the priority. The dashboard is published with a shareable link and I can walk you through it in a live demo."